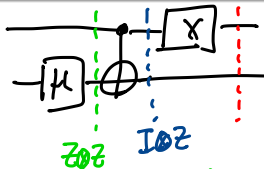


EECE 571Q

Lecture 07: Everything can go wrong

Tuesday 2 June 2026

Announcements



- Quiz 3 today
- Project topic proposals due next Friday 12 June
- Let me know if any of the following presentation dates in July do not work for your team: 7/9, 14/16, 21/23, 28/30

$$(I \otimes H)(Z \otimes X)(I \otimes H) = Z \otimes (HXH) = Z \otimes Z$$

$$(CNOT)(Z \otimes Z)(CNOT) = I \otimes Z$$

$$(X \otimes I)(I \otimes Z)(X \otimes I) = I \otimes Z$$

Last time

We proved the quantum Hamming bound.

→ what about degenerate??

If a non-degenerate code that corrects up to t errors (on n physical qubits and k logical qubits) exists,

$$\left(\sum_{j=0}^t \binom{n}{j} 3^j \right) \cdot 2^k \leq 2^n$$

of ways to have t errors
logical space needed
physical space

Parameters of the smallest code with $k = 1$, $t = 1$: $[[5, 1, 3]]$

Last time

We defined the *Clifford group*.

(Source: Gottesman Def. 6.1)

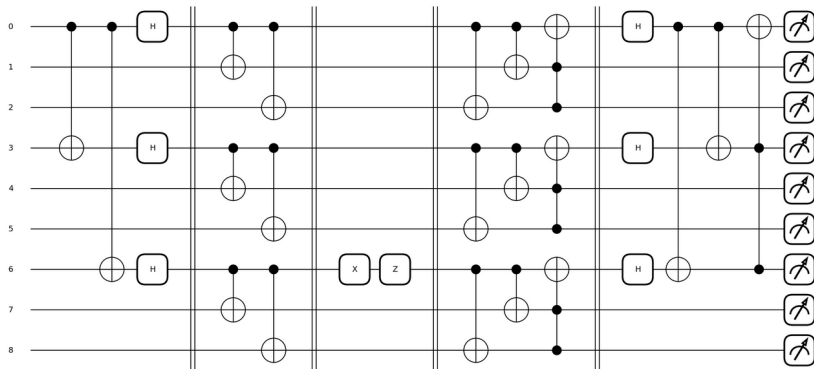
“The Clifford group $\mathcal{C}_n$ on n qubits is the normalizer of $\mathcal{P}_n$ in the unitary group $\mathcal{U}(2^n)$ ”

$$\mathcal{C}_n = \left\{ U \in \mathcal{U}(2^n) \mid U P U^\dagger \in \mathcal{P}_n \ \forall P \in \mathcal{P}_n \right\}$$

Clifford operations map stabilizer codes to other stabilizer codes.
They can be classically simulated in polynomial time.

Guiding question

How can doing all these extra operations be *better* if they can *all* experience noise?

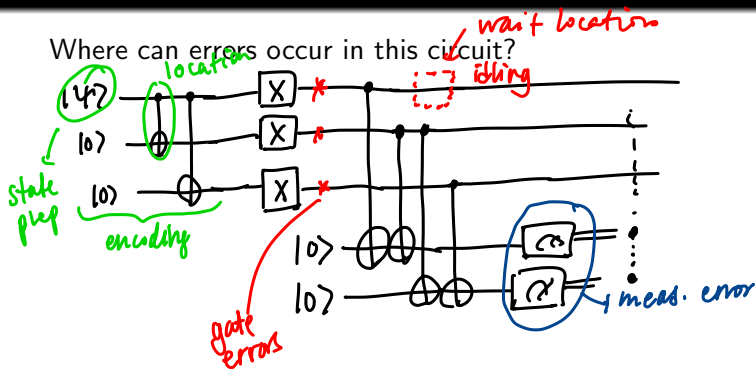


Learning outcomes

Part 2: fault-tolerant quantum computing

- 1 Define and give examples of errors in three types of noise model: code capacity, phenomenological, and circuit-level
- 2 Explain the difference between error correction and fault-tolerance
- 3 State the threshold conditions of a code, and concatenate codes to reduce failure probability

The noise model hierarchy



Code capacity model: errors on data qubits only

Phenomenological noise model: errors on data qubits and on measurement values

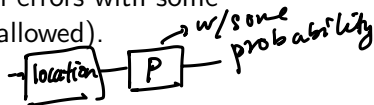
Circuit-level noise: everything (including syndrome measurement circuits) is noisy

Error propagation

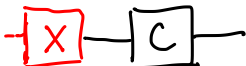
Must consider how errors *propagate* through a circuit.

independent

Assume noise consists of single-qubit Pauli errors with some associated probabilities (some correlation allowed).

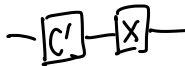


Example 1:



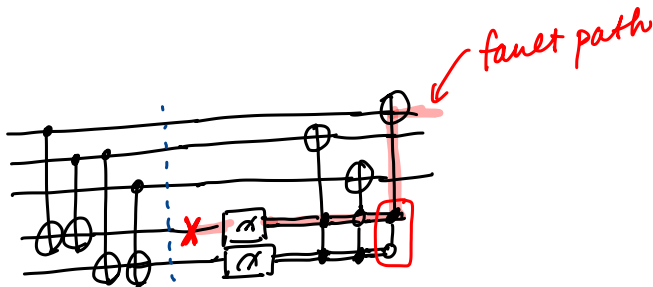
Matrix level:

$$\begin{aligned}
 CX &= (X X^\dagger) C X \\
 &= X \underbrace{(X^\dagger C X)}_{C'}
 \end{aligned}$$



Error propagation

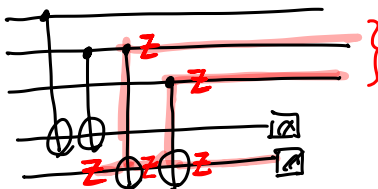
Example 2:



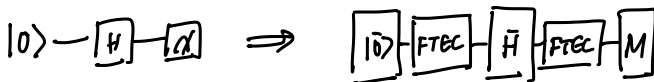
Error propagation

$$(CNOT)(I \otimes Z) CNOT = Z \otimes Z$$

Example 3:



Fault-tolerant simulation



Main idea:

- replace each circuit location with a *fault-tolerant gadget* (implicit: encoded in a QECC)
- add fault-tolerant error correction after the gadgets at each location (except final measurement)

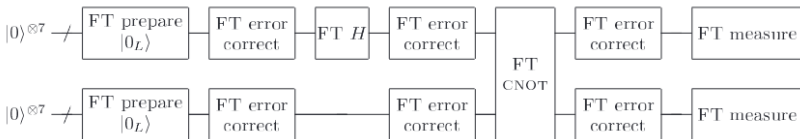
A FT protocol consists of a QECC Q , along with:

- state preparation gadget for at least one state
- meas. gadget for at least one type of meas.
- gate gadgets for universal set of gates
- error correction gadget
- storage gadget

(gadgets act on encoded blocks)

Fault-tolerance: high-level definition

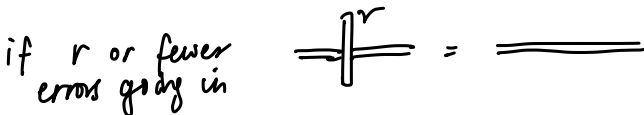
Definition (adapted from Nielsen & Chuang): a procedure is *fault-tolerant* if the failure of a single component leads to *at most* one error per encoded block of qubits in the output.



Fault-tolerance: formal definitions

(Source: Gottesman Def. 10.7)

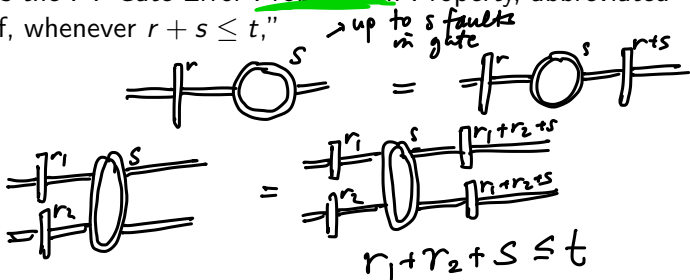
Definition: “The r -filter for the QECC Q is the projector on the subspace spanned by states of the form $E|\psi\rangle$, where $|\psi\rangle \in Q$ and E is an error of weight at most r .”



Fault-tolerance: formal definitions

(Source: Gottesman Def. 10.10)

Definition: “(FT Gate Error Propagation Property). Suppose we have a gate gadget associated with a QECC with distance $2t + 1$ (i.e., it corrects t errors). If it is a single-qubit gate gadget, it satisfies the FT Gate Error Propagation Property, abbreviated GPP, if, whenever $r + s \leq t$,”



Fault-tolerance: formal definitions

(Source: Gottesman Def. 10.8)

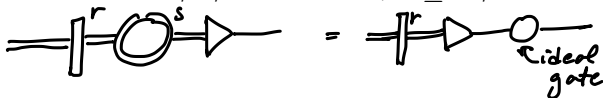
Definition: “The *ideal decoder* for the QECC Q is the quantum channel that takes a state (possibly with errors) encoded in Q , corrects the errors, and then decodes the logical state, discarding all ancilla qubits.”



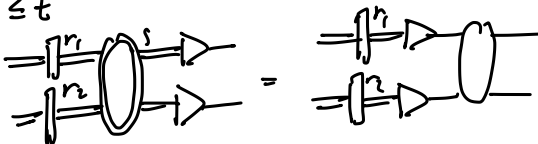
Fault-tolerance: formal definitions

(Source: Gottesman Def. 10.11)

Definition: “(FT Gate Correctness Property). Suppose we have a gate gadget associated with a QECC with distance $2t + 1$. If it is a single-qubit gate gadget, it satisfies the FT Gate Correctness Property, abbreviated GCP, if, whenever $r + s \leq t$,”



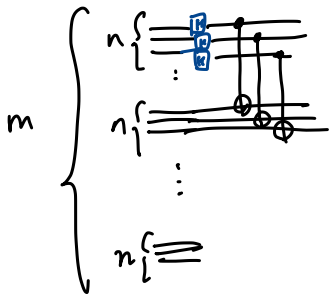
$$r_1 + r_2 + s \leq t$$



Fault-tolerance: transversal gates

Transversal gates automatically satisfy the GPP and GCP.

(Source: Gottesman Def. 11.1)



$$F = \bigotimes_{i=1}^n G_i$$

$\rightarrow G_i$ action on 2^m dim system.

Fault-tolerance: formal definitions

State preparation and measurement

(Source: Gottesman Def. 10.13-15.)

FT preparation error propagation $\text{CNOT}^S = \text{CNOT}^S \text{CNOT}^S$ $(SS \neq t)$

FT preparation correctness $\text{CNOT} \rightarrow \text{CNOT} = \text{CNOT}$ ideal prep.

FT measurement correctness
 $r + SS \neq t$

$$\text{CNOT}^r \text{CNOT}^S = \text{CNOT}^r \text{CNOT} \text{CNOT}^S$$

FT measurement if
equiv. to result of
correctly measuring the correctly
decoded state

Fault-tolerance: formal definitions

Error correction gadgets

(Source: Gottesman Def. 10.17-18)

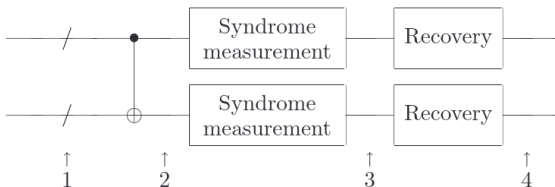
FT error correction recovery $\begin{array}{c} \boxed{\text{FTEC}}^s \\ \hline \hline \end{array} = \begin{array}{c} \boxed{\text{FTEC}}^s \\ \hline \hline \end{array} \begin{array}{c} s \\ \hline \hline \end{array} \begin{array}{c} s \\ \hline \hline \end{array}$

$s \leq t$

FT error correction correctness $\begin{array}{c} \hline \hline \end{array}^r \begin{array}{c} \boxed{\text{FTEC}}^s \\ \hline \hline \end{array} \begin{array}{c} s \\ \hline \hline \end{array} \begin{array}{c} \hline \hline \end{array} = \begin{array}{c} \hline \hline \end{array}^r \begin{array}{c} \hline \hline \end{array}$

Thresholds

If one error occurs before/within this circuit with prob. p , what is the probability of *two or more* errors in the encoded first qubit at the output?



We will start here on Thursday.

Screenshot: Nielsen & Chuang, chapter 10.6.1 (Fig. 10.21)

Thresholds

Consider a CNOT in the Steane code as an example: syndrome measurement circuit is

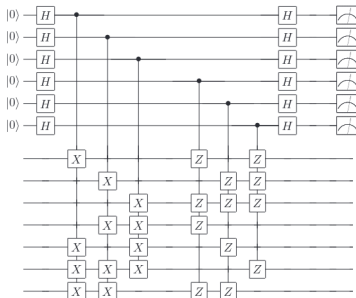


Figure 10.16. Quantum circuit for measuring the generators of the Steane code, to give the error syndrome. The top six qubits are the ancilla used for the measurement, and the bottom seven are the code qubits.

Screenshot: Nielsen & Chuang, chapter 10.5.8.

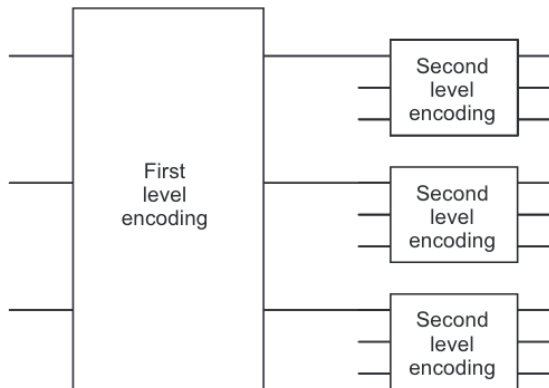
Thresholds

Can work through example of Nielsen & Chuang; key point is that probability of two or more errors in first encoded block is

As long as ϵ^2 , this is less than original error probability
(more robust estimates closer to ϵ)

Concatenation

What if our p is too large? Concatenate codes...



Screenshot: Nielsen & Chuang, chapter 10.6.1 (Fig. 10.22)

Concatenation

Q: if you concatenate k times, what is the failure probability at the k 'th level?

Q: suppose g is the maximum number of operations in a fault-tolerant procedure. What is the simulation overhead?

Q: given a circuit with polynomial $p(n)$ gates, how many times do we need to concatenate to get total accuracy ϵ ?

Threshold conditions

Can find k as long as $p < p_{th} = 1/c$ (*threshold condition*).

The size of the simulation circuit then becomes

Next time

Next class:

- Fault-tolerant simulation with stim
- Estimating the pseudothreshold with simulation

Action items:

- ① Work on project proposal
- ② Download and install stim

Recommended reading:

- For this class: Gottesman Ch. 10, N&C 10.6
- For next class: same as above, plus stim documentation